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WATER COOLING RADIATOR PRESSURE REGULATION STRUCTURE

Abstract

A water cooling radiator pressure regulation structure includes a water cooling radiator and a regulation unit. The water cooling radiator includes a first chamber, a second chamber, and a flow passage unit located between and communicable with the first and the second chamber; and has a cooling liquid contained therein. The regulation unit is communicable with the water cooling radiator, and includes a regulation chamber, a pressure relief section, and a backflow section. The pressure relief section and the backflow section of the regulation unit regulates the cooling liquid when the latter circulates through the regulation unit, relieves pressure caused by thermal expansion, and flows from the regulation unit back to the water cooling radiator.

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Background/Summary

[0001] This application claims the priority benefit of Taiwan patent application number 113108706 filed on Mar. 8, 2024.

FIELD OF THE INVENTION

[0002] The present invention relates to the field of water cooling, and more particularly, to a water cooling radiator pressure regulation structure.

BACKGROUND OF THE INVENTION

[0003] The existing water cooling system includes some necessary components, such as water block, water cooling radiator, water pump and pipes. The water block, the water cooling radiator, and the water pump are serially connected by the pipes to constitute a closed-circuit system. The water block is in contact with a heat source and a cooling liquid in the water block absorbs heat produced by the heat source. The water pump drives the cooling liquid to flow through the pipes and carry the absorbed heat to an outer environment for heat exchanging and reducing the cooling liquid temperature to achieve the purpose of heat dissipation through cooling liquid circulation.

[0004] When the working liquid such as water absorbs heat, it would have an expanded volume but reduced density. The thermally expanded cooling liquid would have a significant influence on the water cooling system that has a fixed volume. For instance, the water block or the water cooling radiator might be pushed by increased liquid pressure in its internal chamber and become deformed, or the joints between the pipes and the water block and/or water cooling radiator might break.

[0005] Following problems will occur when the above conditions happened: [0006] (1) The water block is deformed to result in thermal resistance; [0007] (2) The joints of the pipes and different components in the water cooling system burst to result in leaking of the cooling liquid; and [0008] (3) The water cooling radiator would deform to result in poor heat dissipation and the absorbed heat could not be removed effectively.

[0009] In other words, any one of the above conditions would lead to failure of the whole water cooling system, which in turn causes undesired problems to the electronic device using the water cooling system, such as crash or burn-out of the electronic device due to overheat or short-circuit due to leakage of the cooling liquid.

[0010] Presently, to solve the above problems, a water cooling system with a pressure relief valve provided therein is developed. When the cooling liquid is rise in temperature and accordingly produces additional pressure in the water cooling system, the pressure relief valve opens automatically to balance the internal pressure of the water cooling system. However, the provision of the pressure relief valve causes another problem, i.e. the cooling liquid in the water cooling system will decrease substantially during pressure relief. Since the decreased cooling liquid in turn results in lowered efficiency of the water cooling system, it is necessary to replenish the cooling liquid now and then to maintain the efficiency of the water cooling system.

[0011] It is therefore tried by the inventor to develop a water cooling radiator pressure regulation structure to overcome the above problems.

SUMMARY OF THE INVENTION

[0012] To overcome the problems in the prior art water cooling system, a primary object of the present invention is to provide a water cooling radiator pressure regulation structure for a water cooling system to regulate automatically the pressure of a cooling liquid circulating through the water cooling system.

[0013] Another object of the present invention is to provide a water cooling radiator pressure regulation structure, which enables a cooling liquid to flow back in a water cooling system automatically.

[0014] To achieve the above and other objects, the water cooling radiator pressure regulation structure according to the present invention includes a water cooling radiator and a regulation unit. The water cooling radiator includes a first chamber, a second chamber, and a flow passage unit located between and communicable with the first and the second chamber; and has a cooling liquid contained therein. The regulation unit is communicable with the water cooling radiator, and includes a pressure relief section and a backflow section. The pressure relief section and the backflow section of the regulation unit regulate the cooling liquid when the latter circulates through the regulation unit, relieves pressure caused by thermal expansion, and flows from the regulation unit back to the water cooling radiator.

[0015] With the above structure, when the cooling liquid in the water cooling system is rise in temperature, a part of the cooling liquid would enter a regulation chamber of the regulation unit via the pressure relief section automatically. On the other hand, when the cooling liquid is dropped in temperature, the cooling liquid that is in the regulation chamber would flow from the regulation chamber back into the water cooling radiator via the backflow section, such that the pressure in the water cooling system is regulated automatically.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The structure and the technical means adopted by the present invention to achieve the above and other objects can be best understood by referring to the following detailed description of the preferred embodiments and the accompanying drawings, wherein

[0017] FIG. 1A is an exploded perspective view of a water cooling radiator pressure regulation structure according to a preferred embodiment of the present invention;

[0018] FIG. 1B is an assembled view of FIG. 1A;

[0019] FIG. 1C is a horizontal sectional view of FIG. 1B;

[0020] FIG. 1D is a vertical sectional view of FIG. 1B;

[0021] FIG. 2A is an end sectional view showing how the water cooling radiator pressure regulation structure of the present invention works when a cooling liquid therein is rise in temperature;

[0022] FIG. 2B is another end sectional view showing how the water cooling radiator pressure regulation structure of the present invention works when the cooling liquid therein is dropped in temperature; and

[0023] FIG. 3 shows a water cooling system using the water cooling radiator pressure regulation structure of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0024] The present invention will now be described with some preferred embodiments thereof. For the purpose of easy to understand, elements that are the same in the preferred embodiments are denoted by the same reference numerals.

[0025] Please refer to FIGS. 1A and 1B, which are exploded and assembled perspective views, respectively, of a water cooling radiator pressure regulation structure according to a preferred embodiment of the present invention; and to FIGS. 1C and 1D, which are horizontal and vertical sectional views, respectively, of FIG. 1B. As shown, the water cooling radiator pressure regulation structure of present invention includes a water cooling radiator **1** and a regulation unit **2**. The water cooling radiator **1** mainly includes a first chamber **11**, a second chamber **12**, and a flow passage unit **13**. The flow passage unit **13** is located between the first chamber **11** and the second chamber **12** and communicable with the two chambers **11**, **12**. Further, the water cooling radiator **1** internally contains a cooling liquid.

[0026] The first chamber **11** has a liquid inlet **111**, a liquid outlet **112**, and a partitioning board **113**

dividing an internal space of the first chamber **11** into a liquid-in compartment **114** and a liquid-out compartment **115**. The liquid inlet **111** is communicably connected to the liquid-in compartment **114**, and the liquid outlet **112** is communicably connected to the liquid-out compartment **115**.

[0027] The flow passage unit **13** includes a plurality of pipes **131** and a plurality of radiating fins **132**. The pipes **131** are connected at their respective first end **131a** to the first chamber **11**, and at their respective second end **131b** to the second chamber **12**. The pipes **131** are parallelly arranged with the radiating fins **132** located between any two adjacent pipes **13**.

[0028] The regulation unit **2** includes a first case member **21** and a second case member **22**. The first and the second case member **21**, **22** are connected with each other to together define a regulation chamber **23** between them. The second case member **22** are provided with a pressure relief section **221** having a first check valve assembly **24** mounted thereto and a backflow section **222** having a second check valve assembly **25** mounted thereto. The first check valve assembly **24** includes a lower inlet **241** and an upper outlet **242**; and the second check valve assembly **25**

includes an upper inlet **251** and a lower outlet **252**. The first case member **21** is provided with a through bore **211** for receiving a sealing element **212**, such as a stopper, a spigot, or a plug therein.

[0029] The regulation unit **2** is communicable with the water cooling radiator **1**. In the illustrated preferred embodiment, the regulation unit **2** is connected to a top of the second chamber **12**. The first check valve assembly **24** is communicably connected at the lower inlet **241** to the second chamber **12**, and at the upper outlet **242** to the pressure relief section **221** of the regulation unit **2**. The second check valve assembly **25** is communicably connected at the lower outlet **252** to the second chamber **12**, and at the upper inlet **251** to the backflow section **222** of the regulation unit **2**.

[0030] The arrows in the drawings indicate flow directions of the cooling liquid. Please refer to FIGS. **1A**, **1C** and **1D**. The cooling liquid enters the liquid-in compartment **114** via the liquid inlet **111** and then flows from the first ends **131a** of the pipes **131** that are located at a first side in the flow passage unit **13** (i.e. the left side when viewing in front of the drawing) toward the second ends **131b** of the same pipes **131** into the second chamber **12**. Thereafter, the cooling liquid in the second chamber **12** flows from the second ends **131b** of the pipes **131** that are located at a second side (or a right side) in the flow passage unit **13** toward the first ends **131a** of the same pipes **131** into the liquid-out compartment **115**, from where the cooling liquid leaves the water cooling radiator **1** via the liquid outlet **112**.

[0031] FIGS. **2A** and **2B** are sectional views showing how the water cooling radiator pressure regulation structure of the present invention works, and FIG. **3** shows a water cooling system using the water cooling radiator pressure regulation structure of the present invention. Please refer to FIGS. **2A**, **2B** and **3** along with FIGS. **1A** to **1D**. In the water cooling radiator pressure regulation structure of the present invention, the liquid inlet **111** and the liquid outlet **112** of the water cooling radiator **1** are connected to an end of pipes **3a** and **3b**, respectively; while another end of the pipes **3a**, **3b** are connected to a water block **4**. A set of protection covers **14** (see FIG. **1A**) is provided on an outer side of the flow passage unit **13** of the water cooling radiator **1** and a fan **15** is mounted on a top of the protection covers **14** (see FIG. **3**) to form a complete water cooling system. Please refer to FIG. **2A**, which shows the state in the water cooling radiator pressure regulation structure of the present invention when the cooling liquid is rise in temperature. It is noted the cooling liquid in a temperature raised condition would expand to have reduced density and generate increased pressure. As a result, the expanded cooling liquid will pass the lower inlet **241** to push against a first valve **243** of the first check valve assembly **24** and pass the lower outlet **252** to push against a second valve **253** of the second check valve assembly **25**. The first valve **243** subjected to the pressure pushes a first elastic element **244** upward in the first check valve assembly **24** and accordingly, a gap is formed in the first check valve assembly **24**. With the gap, the cooling liquid is allowed to flow from the second chamber **12** through the gap in the first check valve assembly **24** into the regulation chamber **23**. Meanwhile, the second valve **253** subjected to the pressure is fully attached to and seals a sleeve of the second check valve assembly **25**, such that the cooling liquid

can enter the regulation chamber **23** only via the first check valve assembly **24**.

[0032] Please refer to FIG. **2B**, which shows the state in the water cooling radiator pressure regulation structure of the present invention when the cooling liquid is dropped in temperature. It is noted that the cooling liquid in a temperature lowered condition would restore from expanded volume and reduced density to a normal volume and density, and accordingly generate less pressure. At this point, the first valve **243** and the second valve **253** would be forced to move toward the second chamber **12** due to the weight of the cooling liquid in the regulation chamber **2** and the reduced pressure in the second chamber **12**. Meanwhile, the first elastic element **244** of the first check valve assembly **24** also elastically restores to its original position to push against the first valve **243**, bringing the first valve **243** to fully contact with and accordingly seal a sleeve of the first check valve assembly **24**. On the other hand, the second valve **253** moves naturally toward the second chamber **12** under gravitational force, which results in the formation of a gap between the sleeve and the second check valve **253** of the second check valve assembly **25**. Therefore, the cooling liquid can flow from the regulation chamber **23** of the regulation unit **2** through the gap in the second check valve assembly **25** into the second chamber **12**.

[0033] The position of the first valve **243** and the second valve **253** changes (i.e. rises or sinks) with the temperature of the cooling liquid. Since the second chamber **12** and the regulation unit **2** communicate with each other via the pressure relief section **221** and the backflow section **222**, the additional volume of the expanded cooling liquid in the second chamber **12** resulted from the rising temperature thereof would flow from the second chamber **12** into the regulation chamber **23** via the pressure relief section **221**; and the cooling liquid stayed in the regulation chamber **23** would flow through the backflow section **222** into the second chamber **12** again when the temperature of the cooling liquid drops. In this manner, the cooling liquid is regulated when it circulates through the regulation chamber **23**, relieves pressure caused by thermal expansion, and flows backward into the second chamber **12**.

[0034] The sealing element **212** can be removed to expose the through bore **211** for filling the cooling liquid at the first use of the whole water cooling system or when it is necessary to replenish or replace the cooling liquid. The cooling liquid is filled into the regulation unit **2** via the through bore **211**. When the regulation unit **2** is fully filled with the cooling liquid, any cooling liquid being further filled into the regulation unit **2** would impose a force to push open the second valve **253** of the second check valve assembly **25** and flow into the second chamber **12** to circulate through the whole water cooling system.

[0035] In the illustrated preferred embodiment, the regulation unit **2** is connected to the top of the second chamber **12**. However, in other alternative embodiments, the regulation unit **2** may be otherwise connected to a top of the first chamber **11** or straddled between the first and the second chamber **11**, **12**. In the case of being disposed on the top of the first chamber **11**, the regulation unit **2** works just in the same principle as it is located above the second chamber **12** without being influenced by the difference in its position. For the cooling liquid to flow back under the gravitational force in this case, the liquid inlet **111** and the liquid outlet **112** are changed to two sides of the first chamber **11**. When the regulation unit **2** is straddled between the first and the second chamber **11**, **12**, the first check valve assembly **24** and the second check valve assembly **25** are not particularly limited to be communicable with a specific one of the first and the second chamber **11**, **12**, respectively. For example, in one of many alternative embodiments, the first check valve assembly **24** is communicable with the first chamber **11** while the second check valve assembly **25** is communicable with the second chamber **12**. However, it is understood the present invention is not limited to the above embodiments. In a further embodiment, the first check valve assembly **24** is communicable with the second chamber **12** while the second check valve assembly **25** is communicable with the first chamber **11**.

[0036] In the above cooling liquid circulation process, no additional operation is needed. The pressure of the cooling liquid in the water cooling radiation **1** is automatically regulated with the

change in the temperature of the heat source to avoid undesirable problems or damages of the water cooling system due to overpressure.

[0037] Briefly, the water cooling radiator pressure regulation structure of the present invention has the following advantages: [0038] (1) It is able to regulate the pressure of the cooling liquid automatically; [0039] (2) It does not need replenishment of the cooling liquid frequently; and [0040] (3) It has simple structure.

[0041] The present invention has been described with some preferred embodiments thereof and it is understood that many changes and modifications in the described embodiments can be carried out without departing from the scope and the spirit of the invention that is intended to be limited only by the appended claims.

Claims

1. A water cooling radiator pressure regulation structure, comprising: a water cooling radiator including a first chamber, a second chamber, and a flow passage unit located between and communicable with the first and the second chamber; and the water cooling radiator internally containing a cooling liquid; and a regulation unit being communicable with the water cooling radiator and including a pressure relief section and a backflow section for regulating the cooling liquid when the latter circulates through the regulation unit, relieves pressure caused by thermal expansion, and flows from the regulation unit back to the water cooling radiator.
 2. The water cooling radiator pressure regulation structure as claimed in claim 1, wherein the regulation unit includes a first case member and a second case member, which are connected with each other to together define a regulation chamber between them; the pressure relief section and the backflow section being provided on the second case member; the pressure relief section having a first check valve assembly mounted thereto; and the backflow section having a second check valve assembly mounted thereto.
 3. The water cooling radiator pressure regulation structure as claimed in claim 2, wherein the first check valve assembly has a lower inlet and an upper outlet, the lower inlet being communicable with the second chamber, and the upper outlet being communicable with the regulation unit.
 4. The water cooling radiator pressure regulation structure as claimed in claim 2, wherein the second check valve assembly has an upper inlet and a lower outlet, the upper inlet being communicable with the regulation unit, and the lower outlet being communicable with the second chamber.
 5. The water cooling radiator pressure regulation structure as claimed in claim 2, wherein the first case member is provided with a through bore, in which a sealing element is removably fitted.
 6. The water cooling radiator pressure regulation structure as claimed in claim 1, wherein the first chamber is provided with a liquid inlet, a liquid outlet, and a partitioning board dividing an internal space of the first chamber into a liquid-in compartment and a liquid-out compartment; and the liquid inlet and the liquid outlet being communicably connected to the liquid-in compartment and the liquid-out compartment, respectively.
 7. The water cooling radiator pressure regulation structure as claimed in claim 1, wherein the flow passage unit includes a plurality of pipes and a plurality of radiating fins; the pipes having their respective first end connected to the first chamber and their respective second end connected to the second chamber; and the radiating fins being located between any two adjacent pipes.
 8. The water cooling radiator pressure regulation structure as claimed in claim 1, wherein the flow passage unit includes a set of protection covers, onto which a fan is mounted.
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